

Solution Requirements

Date	05 JULY 2026
Team ID	SWTID-2026-2423
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>[Describe how users will securely log in or verify their identity]</i>	High
2	Authorization levels	<i>[Define the user roles and what access/permissions each role has]</i>	High
3	External interfaces	<i>[Detail any third-party APIs, external databases, or</i>	Medium

		<i>services the system must connect to]</i>	High
4	Transactions processing	<i>[Explain how data exchanges, payments, or core system transactions will be handled]</i>	High
5	Reporting	<i>[Describe the types of reports, dashboards, or data exports the system must generate]</i>	High
6	Business rules, etc.	<i>[List specific logic, constraints, or workflows unique to the business process]</i>	High
7	Compliance to laws or regulations	<i>[Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)]</i>	High
8	<i>Other</i>	<i>[Add any additional functional requirements here]</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>Fast text sentiment extraction and page rendering.</i>	<i>Page loads and classifications take < 1.5 seconds.</i>
2	Scalability	<i>Handles high traffic during peak exams.</i>	<i>Supports up to 50,000 concurrent active users</i>
3	Security & Data Privacy	<i>Protects student text data and identity tokens.</i>	<i>End-to-end AES-256 and TLS 1.3 data encryption.</i>
4	Reliability & Availability	<i>System remains active for night-time learners.</i>	<i>Minimum of 99.9% system uptime per month.</i>
5	Usability & Accessibility	<i>Clear layouts compatible with accessibility tools.</i>	<i>Full compliance with WCAG 2.1 AA standards.</i>
6	<i>Other</i>	<i>Allows hot-swapping AI models without downtime.</i>	Maintenance windows under 2 hours per month.